

BANARAS HINDU UNIVERSITY

B.Sc. (Hons) Semester III Mid-Semester Examination 2025-26 (NEP SEC)

SEC — CHEMISTRY**Paper Code: CHESE31 / CHESE21 : Synthetic Methodologies**

Time: 1 Hour

Maximum Marks: 20

1. Answer the following:

- (a) Describe the basic principle of distillation of organic solvents with suitable equations and examples. Draw the schematic diagram of solvent distillation assembly used in the chemical laboratory. [4]
- (b) What are molecular sieves? What are their approximate chemical compositions? Write down the classification of molecular sieves with specific examples of their use. [4]
- (c) Describe the drying methodologies for THF and Methanol in detail. [2]

2. Answer the following:

- (a) What are the key characteristics that define a click reaction? Explain the mechanism of the Copper(I)-catalyzed azide–alkyne cycloaddition (CuAAC) reaction. [4]
- (b) Describe the thiol–ene click reaction and its mechanism. [3]
- (c) Predict the correct products for the following reactions: [3]
- (1) Phenyl azide + Phenylacetylene $\xrightarrow{\text{Cu(I)}}$
- (2) 2-Methylbut-3-yn-2-ol + Phenyl azide $\xrightarrow{\text{Cp}^*\text{RuCl}(\text{PPh}_3)_2}$